EBIN BABU THOMAS

AI Engineer: LLM Agents, RAG and Evaluation

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Mavelikara, Kerala, India · Remote · Open to relocation: Bengaluru / Hyderabad

SUMMARY

AI engineer who builds LLM agents, search systems and the tests that check them. Shipped Diarist, a local meeting transcriber now in public beta with about 300 downloads. Built hybrid search over 433 city council meetings, and agent test harnesses that run across Anthropic, OpenAI, Gemini and open-weight models. Before that, 3.5 years building LLM, RAG and backend systems for startup clients in the US, Canada, Europe and Australia.

TECHNICAL SKILLS

LLM applications: Python, FastAPI, LLM agents and tool use (Claude Agent SDK, Anthropic API, Google Gemini API, OpenAI API, PydanticAI, LangGraph, LangChain), MCP (FastMCP), hybrid RAG (BM25, vector search, RRF, cross-encoder reranking), prompt and context engineering, structured outputs, Playwright

Models and data: PyTorch, Hugging Face Transformers / PEFT, fine-tuning (LoRA / QLoRA, DPO), LLM serving (vLLM, Modal), vector databases (LanceDB, Qdrant), PostgreSQL / SQL, SQLite (FTS), MongoDB, speech (Whisper, NVIDIA Nemotron ASR and diarization), OCR and document processing (RapidOCR, PDFium, PyMuPDF, OpenCV)

Evaluation: eval harness design (Inspect), retrieval benchmarking (nDCG, recall@k), agent test sweeps in Docker sandboxes, LLM-as-judge with human agreement checks, bootstrap and permutation statistics, cost and latency tracking, Langfuse, OpenTelemetry

Engineering: Docker, Git, GitHub Actions CI/CD, pytest, uv, Linux, AWS, GCP (Cloud Run, GKE), Terraform, OAuth2 / JWT; familiar with Rust / Tauri 2, Svelte 5, TypeScript, Next.js

EXPERIENCE

Software Engineer · Zackriya Solutions *Jan 2022 – Jun 2025*

Remote, India. Services firm building first versions of products, AI systems and data tools for startup clients in the US, Canada, Europe and Australia. Grew from trainee to core developer; owned client backends end to end and scoped incoming projects.

- **FinBot, financial RAG and fine-tuned LLM:** fine-tuned Mistral 7B (QLoRA) and built a streaming news pipeline (Alpaca news API → Bytewax → Qdrant) so answers cite live market news; served with vLLM on a GKE GPU node.
- **Natural-language property search:** built a plain-English-to-SQL search backend on FastAPI (GPT-3.5 query translation with "Did you mean?" suggestions) for real-estate listings; load-tested and deployed on Google Cloud Run.
- **DocuAI, semantic document retrieval:** designed the chunking, ingestion and Qdrant vector-search pipeline over 1,000+ documents with a Next.js front end; delivered and demoed the working prototype in two days.
- **Meetily, the company's open-source meeting-notes app:** refactored the backend and added OpenAI provider support (PR #75).
- **Data and speech pipelines:** sensor-data ingestion (MQTT → MongoDB) for a bioreactor startup; spoken-English scoring from short videos (Whisper, Microsoft's pronunciation API, AWS Fargate); healthcare-device compliance pipelines.

PROJECTS

Diarist: Local Meeting Transcriber with Speaker Labels, for Windows *Oct 2026 – now*

Public beta, six releases, about 300 downloads; free and MIT-licensed. diarist.dev · github.com/ebt55/diarist (NVIDIA NeMo-Speech.cpp (CUDA), Nemotron speech and speaker models, Svelte 5, TypeScript, SQLite (FTS))

- A Windows app that records a meeting, or imports a recording, and writes a transcript with a speaker label on every line. It runs on the user's own GPU: no cloud, no account, no Python install.
- The speech models are NVIDIA's; I built everything around them: audio capture, echo handling, the model manager, the transcript database and editor, the installer and updater. An hour of audio takes about two minutes on an RTX 3080.
- Every release must pass fixed accuracy and speed checks on 28 hours of recordings. **I turned down a 40% faster runtime** because it missed two of them. Reported two bugs in NVIDIA's runtime; both were fixed upstream.

Council-meeting search and a retrieval benchmark *Oct 2026*

Two systems from a 48-hour take-home (private repos). (Python, LanceDB (BM25 + vector + RRF), cross-encoder reranking, Qwen3-4B, vLLM, SigLIP, RapidOCR)

- Hybrid search over captions for **433 Seattle City Council meetings**, about 813 hours, with a lookup tool over the official vote record. Answers cite the video to the second.
- The evaluation showed where it works: the record lookup answers vote questions, caption search does not yet answer open questions well, and larger models did not change that. A speaker-labelled transcript recovered 17 of 18 roll-call votes the live captions had skipped.

- The benchmark stores evidence as time spans, so retrievers with different chunking are scored against the same ground truth, and tests whether its scores predict what a reader model answers. My own proposed metric did no better than nDCG, and the write-up says so.

ProofPack: Web-Checking Agent That Must Show Proof for Each Finding

Jul – Sep 2026

Built from a challenge brief. A project, not a product; no users. github.com/ebt55/proofpack (Python, Gemini API, Playwright, FastAPI, Jinja2, pytest)

- A Gemini agent reads a purchase request and checks the provider's public website the way a compliance reviewer would.
- The model only proposes findings. Plain code **rejects any finding without a saved, hashed screenshot** and a quote that is really on a page the agent visited, and leaves every approve or deny decision to a person. The rejection paths are unit-tested without a model.

Kobayashi Maru: Test Harness for Rule-Breaking by Coding Agents

Sep 2026

Apart Research and CeSIA AI Incident Response sprint, solo. github.com/ebt55/kobayashi-maru (Python, Docker, Anthropic API, OpenRouter, Ollama, statsmodels)

- A harness that runs coding agents in Docker sandboxes across four model providers, detects rule-breaking with plain code, and re-grades every submission in a clean container. 8,959 runs on six models, resumable after failures, with cost tracking.
- Used it to test whether impossible tasks make agents cheat on solvable ones. On one model the habit **spread through the agent's own notes** and stopped when the notes were withheld.

printready: Checks and Fixes AI Sticker Art Against a Printer's Rules *Oct 2026*

Work sample; rules from Sticker Mule's public FAQ, not affiliated. github.com/ebt55/printready (Python, OpenCV, scikit-image, Pillow, xAI API, MCP server, pytest)

- Checks AI-made sticker art before printing, removes plain backgrounds, adds the border and cut line, and sends what it cannot fix safely to a person. A guard undoes any fix that changes the art.
- I fixed the test and the keep-or-drop rule before running it. **Letting an AI agent choose the fixes did no better** than the fixed pipeline, at about four times the cost, so the agent is off by default.

One Query Away: Do AI Monitors Check Before They Accuse? *Nov 2026*

Apart Research Swarm Alignment Hackathon. github.com/ebt55/one-query-away (Python, SQLite, Anthropic API, OpenAI API, DeepSeek API, structured outputs)

- Tool-calling monitors from three providers review an agent's log and can query a database before giving a verdict. Built the scenario generator, the query API, per-call logging and spend caps. In a rerun with one query per turn, Claude monitors looked up the record that clears a compliant agent and rarely accused; two GPT monitors still accused nearly all such agents. Synthetic cases.

ExactDoc: PDF to Editable Word Documents, Checked Page by Page *Aug – Oct 2026*

Beta 0.3.0b3 on PyPI. github.com/ebt55/exactdoc (Python, PDFium / pypdfium2, OOXML, LibreOffice headless, PyMuPDF, pytest)

- Converts a PDF into a Word document you can edit, with real headings, tables and paragraphs that re-wrap. It can also write a version made for Google Docs.
- It opens its own output, **compares every page with the PDF** and corrects page breaks and spacing. Rule-based, with a test suite that runs on every change. Designed layouts and equations do not convert well yet.

LLM Pricing Hub Remedies: Do Antitrust Rules Stop a Shared AI Pricing Advisor? *Oct 2026*

Apart Research AI Collusion sprint. github.com/ebt55/llm-pricing-hub-remedies (Python, uv, Anthropic API, OpenAI SDK, OpenRouter, SciPy, pandas)

- A multi-agent market simulation across five providers, with spend caps and resumable parallel runs. Seller models differed sharply in how readily they followed a pricing advisor: one accepted 12% of its nudges, two accepted over 70%.

OPEN-SOURCE CONTRIBUTIONS

- AlignTune and SafeTune **(Lexsi Labs' LLM fine-tuning libraries)**: found that their code-grading rewards gave full marks to code that never passed its tests; reported it privately with a patch and tests; Lexsi merged fixes for both that credit the report (PR #40, PR #32).
- NeMo-Speech.cpp **(NVIDIA's speech runtime)**: reported two bugs with reproductions; both confirmed and fixed by the maintainers (#67, #81).

EDUCATION & CERTIFICATIONS

Technical AI Safety course (BlueDot Impact) *Sep – Oct 2026*

BlueDot certificate (Oct 2026)

Deep RL (Udacity Nanodegree) · Machine Learning (Stanford Online) *2020*

Udacity certificate (Jul 2020) · Coursera certificate (Mar 2020)

B.Tech (CSE), undergraduate coursework *2018 – 2020*

APJ Abdul Kalam Technological University · Sree Buddha College of Engineering, Kerala

Career break Jul 2025 – Feb 2026. Independent engineering and research since Mar 2026.